# Objects and editing

A flow subtracts a quantity from one named account and adds it to another, in one unit. Accounts can hold several units; each account/unit pair has its own starting balance and time.

```
{
"name": "Warehouse", "units": "kg",
"opening_amount": "20",
"opening_at": "2026-10-01 00:00:00"
}

{
"flow_id": "22222222-2222-4222-8222-222222222222",
"date": "2026-10-03 09:15:06", "name": "Materials",
"from_account": "Warehouse", "to_account": "Workshop",
"amount": "0.125", "units": "kg", "status": "Complete",
"category": "Materials", "tags": ["Project"],
"notes": "Exact fractional quantity"
}
```

| Representation | Meaning |
|---|---|
| Account / units | Account names identify holders. Unit labels are required and case-sensitive: mW and MW differ. No implicit conversion. Starting balance is signed decimal text, or null when unknown. |
| Amount / flow ID | Exact decimal text, without a cents limit. .125 stores as 0.125; 12.00 stores as 12. UUID preserves edits/imports; Repeat gets a new UUID when added. |
| Date | One Date/Time entry, optional 24-hour time to seconds. Leading zeros optional. Missing month/day = 1; missing hour/minute/second = 0. Date-only means midnight. |
| Category / tags | One category and a list of tag names. Shared tags can connect related transfers. Ten agreed flow fields plus ID. |

## The GUI edits these objects

Default order: Date, Name, From, To, Amount, Units, Status, Category, Tags, Notes. Defaults: today, USD, Complete and Uncategorized. Units uses the learned dropdown with filtering, Enter and X. Accounts has Name, Units, Starting Balance and Starting Date/Time.

Existing records save completed fields on Enter, Tab or leaving them. Ctrl+S finishes existing edits; new drafts wait for Add. Notes: Ctrl+Enter or leaving/closing saves. Invalid edits stay unsaved with a silent error. New clears; Repeat copies every value, including units and date. Delete removes a selected row without confirmation; undo restores its record. Delete inside text edits text.

Row/page changes and blank ledger clicks discard unadded drafts without prompts; undo restores them. Drag titles to move columns; cell editing and Tab follow their visible order. Account Name/Units identify a row; starting balance/date remain editable. Starting Balance enters with the caret at the end.

Filters: From, To, Status, Categories, Tags and Units. Checkboxes select multiple values; hover lists every selection. Filter selections and choice search support session undo/redo. Unit matches keep exact case.

Actual objects and SQL rows: examples/record_objects.json and examples/review.sql. Source walkthrough: docs/model_review.md. Controls and update/export commands: README.md.

# Storage and calculations

Account/category/tag keys use name.strip().casefold(), retaining display spelling. Unit labels keep exact case. Amounts are canonical decimal TEXT; Python Decimal arithmetic uses sufficient precision for every operand.

| Table | Every stored column |
|---|---|
| accounts | account_key, name |
| account_balances | account_key, units, opening_amount, opening_at |
| flows | flow_id, date, from_key, to_key, amount, units, status, name, notes, category_key |
| categories | category_key, name |
| tags | tag_key, name |
| flow_tags | flow_id, tag_key, position |
| audit | audit_id, changed_at, entity, entity_key, action, before_json, after_json |

account_balances is keyed by (account_key, units); each flow references matching endpoint/unit pairs. Lists show pairs used by saved flows, including Void. Unreferenced anchors remain stored for reuse/undo. audit retains before/after history. SQL: flowledger/schema.sql.

## Save, calculate and inspect

GUI draft -> Repository.save_flow() -> validate -> write flow, endpoint/unit anchors, choices, tags and audit atomically. Failed writes roll back. Ctrl+Z/Ctrl+Y reverse records/deletion, drafts, filters, searches, graph controls and all plot/column changes in one session order.

Within its unit, a flow contributes +amount to To and -amount to From. Confirmed includes Complete; Including Pending adds Pending; Including Planned adds Planned. Void contributes zero. Each pair has a confirmed anchor immediately before events at its own timestamp; unknown remains unknown.

Select account/unit pairs to graph. Different units have separate plots sharing time. Combine sums selected accounts within each unit only. Hover gives exact quantity/unit; drawing uses floating-point coordinates. A transfer cancels if both endpoints are selected in its unit. Ledger/category totals stay separate by unit.

## Migration and scope

Opening schema 1-4 first saves .pre-v5-timestamp.bak, then upgrades atomically. Existing cents and anchors become exactly equivalent USD quantities; IDs, metadata and old audit remain. Older forecasts stay in the backup and recovery JSON, outside active flows. Reopening makes no new migration copy.

Import/export: ten fields plus ID; units required. Git rollback restores code, not migrated records; older apps need the pre-v5 backup. Scope: any units, multiple per-unit anchors, local timestamps, one computer. No automatic conversion or forecast collection controls. Executed checks: docs/test_results.txt and docs/units_interaction.txt.